

EXP NO: 8

Client-Side Form Validation for a Library Management System Using JavaScript with Database Integration via Servlet/JSP

AIM:

Consider a Library Management System. Develop a JavaScript program that will validate the controls in the forms you have created for the application. State the assumptions you make (business logic you are taking into consideration). Note: Your application must access a database using Servlet/JSP.

CODE:

library_managemet_system.jsp

```
<%@ page import="java.sql.*" %>
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<html>
<head>
<title>Library Management - Add Book</title>
<style>
body {
    font-family: Arial, sans-serif;
    background-color: #f3f3f3;
    margin: 20px;
    padding: 20px;
}

h2 {
    color: #333;
}

form {
    background-color: #fff;
    padding: 20px;
    border-radius: 8px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    width: 400px;
}

input[type="text"],
input[type="number"],
select {
    width: 100%;
    padding: 8px;
    margin: 6px 0 12px 0;
    border: 1px solid #ccc;
    border-radius: 4px;
}

input[type="submit"] {
    background-color: #4CAF50;
    color: white;
```

```

border: none;
padding: 10px 16px;
cursor: pointer;
border-radius: 4px;
}

input[type="submit"]:hover {
  background-color: #45a049;
}

p {
  font-size: 16px;
}

a {
  color: #007BFF;
  text-decoration: none;
}

a:hover {
  text-decoration: underline;
}
</style>

<script>

function validateForm() {
  const bookId = document.getElementById("bookId").value.trim();
  const title = document.getElementById("title").value.trim();
  const author = document.getElementById("author").value.trim();
  const year = document.getElementById("year").value.trim();
  const category = document.getElementById("category").value;
  const copies = document.getElementById("copies").value.trim();
  const currentYear = new Date().getFullYear();

  if (!/^[a-zA-Z0-9]+$/.test(bookId)) {
    alert("Book ID must be alphanumeric.");
    return false;
  }
  if (!/^[A-Za-z ]+$/.test(title)) {
    alert("Title must contain only letters.");
    return false;
  }
  if (!/^[A-Za-z ]+$/.test(author)) {
    alert("Author name must contain only letters.");
    return false;
  }
  if (!/^\d{4}$/.test(year) || parseInt(year) > currentYear) {
    alert("Year must be a valid 4-digit year.");
    return false;
  }

  if (category === "") {
    alert("Please select a category.");
    return false;
  }
}

```

```

    }
    if (isNaN(copies) || parseInt(copies) <= 0) {
        alert("Copies must be a positive number.");
        return false;
    }

    return true;
}
</script>
</head>
<body>
<h2>Add Book to Library</h2>
<form method="post" onsubmit="return validateForm()">
    Book ID: <input type="text" name="bookId" id="bookId"/><br/>
    Title: <input type="text" name="title" id="title"/><br/>
    Author: <input type="text" name="author" id="author"/><br/>
    Year: <input type="text" name="year" id="year"/><br/>
    Category:
    <select name="category" id="category">
        <option value="">--Select--</option>
        <option value="Science">Science</option>
        <option value="Fiction">Fiction</option>
        <option value="History">History</option>
    </select><br/>
    Copies: <input type="number" name="copies" id="copies"/><br/><br/>
    <input type="submit" value="Add Book"/>

<p>
<label>Book Details:-</label>
<a href="/display_books_79.jsp">Click Here</a>
</p>
</form>

<%

// JSP Database logic

// TODO:Change password before run the code
// TODO: CREATE THE TABLE IN MYSQL
/**
CREATE TABLE books (
    book_id VARCHAR(20) PRIMARY KEY,
    title VARCHAR(100),
    author VARCHAR(100),
    year INT,
    category VARCHAR(50),
    copies INT
);
**/

String bookId = request.getParameter("bookId");
if (bookId != null) {
    String title = request.getParameter("title");
    String author = request.getParameter("author");
    String year = request.getParameter("year");

```

```

String isbn = request.getParameter("isbn");
String category = request.getParameter("category");
String copies = request.getParameter("copies");

try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection conn = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/library_management_karthick", "root", "");
    PreparedStatement ps = conn.prepareStatement(
        "INSERT INTO books (book_id, title, author, year, category, copies) VALUES (?, ?, ?, ?, ?,
?)");
    ps.setString(1, bookId);
    ps.setString(2, title);
    ps.setString(3, author);
    ps.setInt(4, Integer.parseInt(year));
    ps.setString(5, category);
    ps.setInt(6, Integer.parseInt(copies));

    int result = ps.executeUpdate();
    if (result > 0) {
        %>
        <script>alert("Book Added Successfully!!")</script>
        <%
    } else {
        %>
        <script>alert("Failed to add book!!")</script>
        <%
        out.println("<p style='color:red;'>Failed to add book.</p>");
    }

    ps.close();
    conn.close();
} catch (Exception e) {
    out.println("<p style='color:red;'>Error: " + e.getMessage() + "</p>");
}
}
%>
</body>
</html>

```

display_books.jsp

```

<%@ page import="java.sql.*" %>
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<html>
<head>
<title>Library Books List</title>
<style>
table {
    border-collapse: collapse;
    width: 90%;
    margin: 20px auto;
}
th, td {

```

```

border: 1px solid #333;
padding: 10px;
text-align: center;
}
th {
background-color: #f2f2f2;
}
h2 {
text-align: center;
}
</style>
</head>
<body>
<h2>All Books in Library</h2>
<%
try {
Class.forName("com.mysql.cj.jdbc.Driver");
Connection conn = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/library_management_karthick", "root", "");

Statement stmt = conn.createStatement();
ResultSet rs = stmt.executeQuery("SELECT * FROM books");
out.println("<table>");
out.println("<tr><th>Book
ID</th><th>Title</th><th>Author</th><th>Year</th><th>Category</th><th>Copies</th></tr>");

while (rs.next()) {
out.println("<tr>");
out.println("<td>" + rs.getString("book_id") + "</td>");
out.println("<td>" + rs.getString("title") + "</td>");
out.println("<td>" + rs.getString("author") + "</td>");
out.println("<td>" + rs.getInt("year") + "</td>");
out.println("<td>" + rs.getString("category") + "</td>");
out.println("<td>" + rs.getInt("copies") + "</td>");
out.println("</tr>");
}

out.println("</table>");
rs.close();
stmt.close();
conn.close();
} catch (Exception e) {
out.println("<p style='color:red;text-align:center;'>Error: " + e.getMessage() + "</p>");
}
%>
</body>
</html>

```

OUTPUT:

The screenshot shows the Eclipse IDE interface. The Project Explorer on the left lists the project structure, including 'HelloWorld_Rithika' and 'library_management_system_79.jsp'. The main editor displays the HTML code for 'display_books_79.jsp'. The code includes a title 'Library Management - Add Book', a form with a text input, a number input, and a submit button, all styled with CSS.

```

1 <@ page import="java.sql.*" %>
2 <% page contentType="text/html; charset=UTF-8" language="java" %>
3 <html>
4 <head>
5 <title>Library Management - Add Book</title>
6 <style>
7 body {
8     font-family: Arial, sans-serif;
9     background-color: #f3f3f3;
10    margin: 20px;
11    padding: 20px;
12 }
13
14 h2 {
15     color: #333;
16 }
17
18 form {
19     background-color: #fff;
20     padding: 20px;
21     border-radius: 8px;
22     box-shadow: 0 2px 5px rgba(0,0,0,0.1);
23     width: 400px;
24 }
25
26 input[type="text"],
27 input[type="number"] {
28     width: 100%;
29     padding: 8px;
30     margin: 6px 0 12px 0;
31     border: 1px solid #ccc;
32     border-radius: 4px;
33 }
34
35 input[type="submit"] {
36     background-color: #4CAF50;
37     color: white;
38     border: none;
39     padding: 10px 16px;
40     cursor: pointer;
41     border-radius: 4px;
42 }
43
44 input[type="submit"]:hover {
45     background-color: #450049;
46 }
47

```

[illegible]

```

eclipse-workspace - hello_world_mallu/src/main/webapp/library_managemet_system_79.jsp - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer X
> Biju
> Deepak
> geuniv-exp8
> gini
> harini
> haritha
> hello_170
> Hello_world_kumaran
> hello_world_mallu
  > Deployment Descriptor: hello_world_mallu
  > JAX-WS Web Services
  > JRE System Library [jreSE-16]
  > src/main/java
  > Server Runtime [apache-tomcat-9.0.36]
  > Web App Libraries
  > build
  > src
    > main
      > java
        > webapp
          > META-INF
          > WEB-INF
            > display_books_79.jsp
            > index.jsp
            > library_managemet_system_79.jsp
    > HelloWorld_Ritika
    > HelloWorlds
    > indk
    > index_001
    > kabith
    > Krishnamoorthi
    > MinProject_112
    > Ramesh_134
    > Servers
    > srisharan_170
    > vishvam_6

library_managemet_system_79.jsp X display_books_79.jsp
149 p {
150     font-size: 16px;
151 }
152
153 a {
154     color: #007BFF;
155     text-decoration: none;
156 }
157
158 a:hover {
159     text-decoration: underline;
160 }
161 </style>
162
163 <script>
164
165 function validateForm() {
166     const bookId = document.getElementById("bookId").value.trim();
167     const title = document.getElementById("title").value.trim();
168     const author = document.getElementById("author").value.trim();
169     const year = document.getElementById("year").value.trim();
170     const category = document.getElementById("category").value;
171     const copies = document.getElementById("copies").value.trim();
172     const currentYear = new Date().getFullYear();
173
174     if (/^[a-zA-Z0-9]*$/.test(bookId)) {
175         alert("Book ID must be alphanumeric.");
176         return false;
177     }
178     if (/^[A-Za-z ]*$/.test(title)) {
179         alert("Title must contain only letters.");
180         return false;
181     }
182     if (/^[A-Za-z ]*$/.test(author)) {
183         alert("Author name must contain only letters.");
184         return false;
185     }
186     if (/^[d4]$/.test(year) || parseInt(year) > currentYear) {
187         alert("Year must be a valid 4-digit year.");
188         return false;
189     }
190
191     if (category === "") {
192         alert("Please select a category.");
193         return false;
194     }
195     if (isNaN(copies) || parseInt(copies) <= 0) {

```

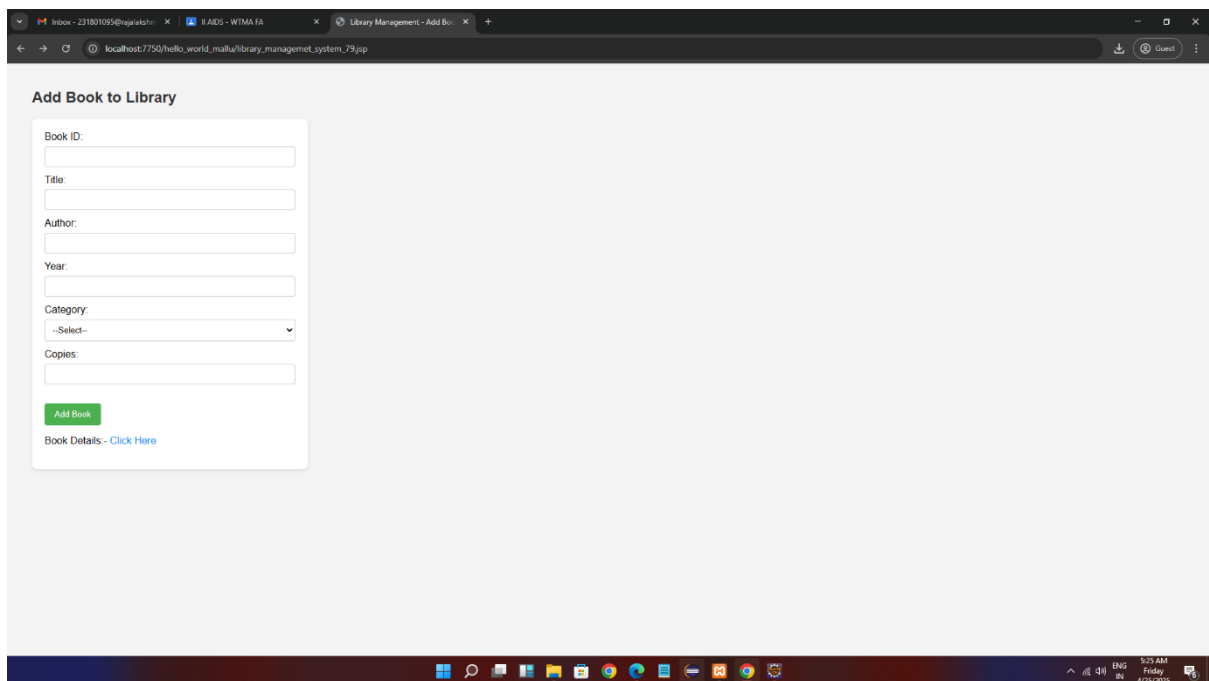
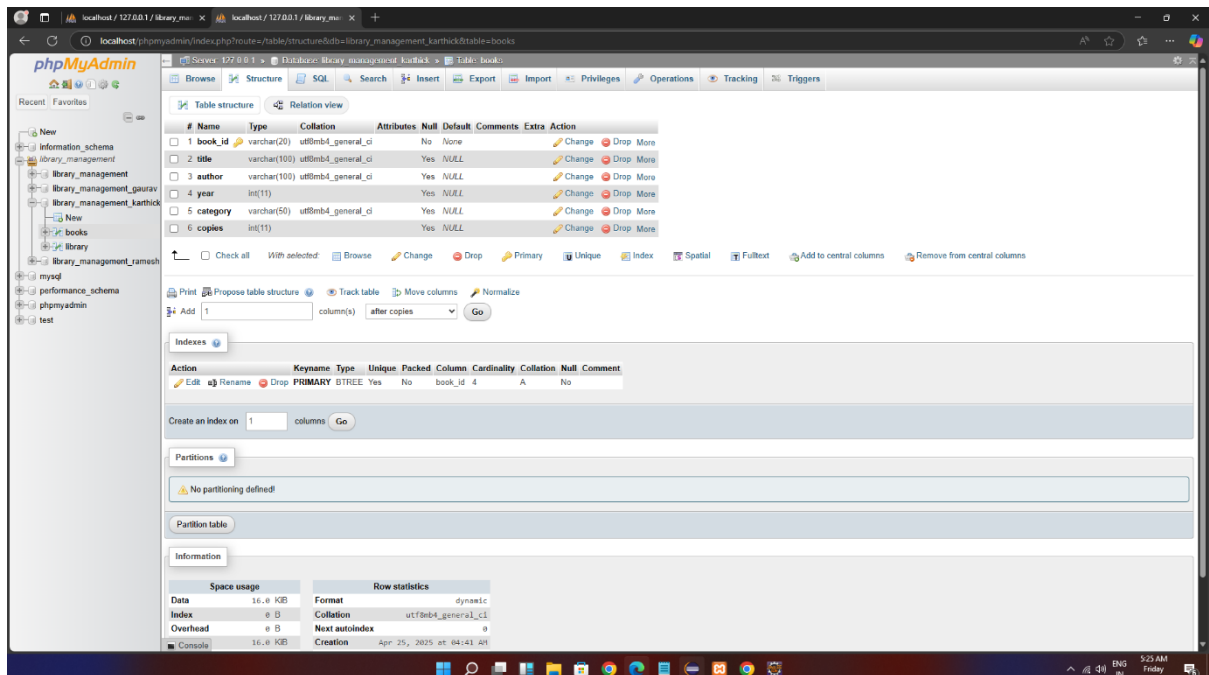
```

eclipse-workspace - hello_world_mallu/src/main/webapp/library_managemet_system_79.jsp - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer X
> Biju
> Deepak
> geuniv-exp8
> gini
> harini
> haritha
> hello_170
> Hello_world_kumaran
> hello_world_mallu
  > Deployment Descriptor: hello_world_mallu
  > JAX-WS Web Services
  > JRE System Library [jreSE-16]
  > src/main/java
  > Server Runtime [apache-tomcat-9.0.36]
  > Web App Libraries
  > build
  > src
    > main
      > java
        > webapp
          > META-INF
          > WEB-INF
            > display_books_79.jsp
            > index.jsp
            > library_managemet_system_79.jsp
    > HelloWorld_Ritika
    > HelloWorlds
    > indk
    > index_001
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    > Krishnamoorthi
    > MinProject_112
    > Ramesh_134
    > Servers
    > srisharan_170
    > vishvam_6

library_managemet_system_79.jsp X display_books_79.jsp
196     alert("Copies must be a positive number.");
197     return false;
198 }
199
200     return true;
201 }
202 </script>
203 </head>
204 <body>
205     <h2>Add Book to Library</h2>
206     <form method="post" onsubmit="return validateForm()">
207         Book ID: <input type="text" name="bookId" id="bookId"/><br>
208         Title: <input type="text" name="title" id="title"/><br>
209         Author: <input type="text" name="author" id="author"/><br>
210         Year: <input type="text" name="year" id="year"/><br>
211         Category:
212         <select name="category" id="category">
213             <option value="">--Select--</option>
214             <option value="Science">Science</option>
215             <option value="Fiction">Fiction</option>
216             <option value="History">History</option>
217         </select><br>
218         Copies: <input type="number" name="copies" id="copies"/><br>
219         <input type="submit" value="Add Book"/>
220
221         <p>
222             <label>Book Details:</label>
223             <a href="/display_books.jsp">Click Here</a>
224         </p>
225     </form>
226
227 <#
228
229 // JSP Database logic
230
231 // TODO: Change password before run the code
232 // TODO: CREATE THE TABLE IN MYSQL
233 /**
234  CREATE TABLE books (
235     book_id VARCHAR(20) PRIMARY KEY,
236     title VARCHAR(100),
237     author VARCHAR(100),
238     year INT,
239     category VARCHAR(50),
240     copies INT

```

Book ID	Title	Author	Year	Category	Copies
001	WTMA	Mallu	2006	Science	95
002	Phy	Karthick	2006	Science	79
003	Crime and Punishment	Fyodor Dostoevsky	1866	Fiction	84
004	No Longer Human	Osamu Dazai	1948	Fiction	79

RESULT:

Thus, form validation using JavaScript and database access through JSP/Servlet was successfully demonstrated by creating a Library Management form.